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Flexible Pay and Labor Supply: Evidence from Uber's Instant Pay

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Abstract. Modern tech platforms provide workers real-time control over when they work and, increasingly, flexible pay: the option to be paid immediately after work. We investigate the labor supply effects of pay flexibility and the implications of present-biased preferences among gig-economy workers. Using granular data from a nationwide randomized controlled trial at Uber, we estimate the effects of switching from a fixed weekly pay schedule to Instant Pay, a system that allows on-demand, within-day withdrawals. We find that flexible pay substantially increased drivers' work time. Furthermore, consistent with present bias, the response is significantly higher when drivers are further away from the end of their counterfactual weekly pay cycle. We discuss welfare and broader implications in contexts in which workers have the ability to flexibly supply labor.

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1. Introduction

Prior to 2020, roughly 30% of the U.S. labor force reported some flexibility over when their workdays started and stopped, a figure that has accelerated with the growth of remote work (Mas and Pallais 2020). Gig-work platforms extend this control further, allowing workers real-time control over their schedules. Whereas this flexibility has obvious benefits for workers, it also introduces the potential for labor supply decisions to reflect many of the same biases and intertemporal patterns commonly found in consumption and savings decisions. That is, when workers can choose their labor supply, the typical structure of up-front costs (effort) and delayed benefits (wages) may result in procrastination and the undersupply of work. As a result, variation in the timing of pay over relatively short horizons may have surprisingly large effects on labor supply.

We use a large-scale natural field experiment at Uber to study the labor supply effects of a key component of the human resources toolkit: the timing/flexibility of pay. Prior to the experiment, Uber—similar to roughly a quarter of employers in the United States (Bureau of Labor Statistics 2023)—paid workers on a weekly cycle. Across most markets, drivers could take any number of rides from Monday at 4 am to the following Monday at 4 am and would be paid their accumulated earnings on the subsequent Wednesday. Thus, rides driven on Sunday would be paid roughly three days later, whereas those on Monday (after 4 am) would be paid roughly nine days later. In March 2016, Uber began to experimentally roll out Instant Pay, an earned wage access (EWA) product that allowed drivers to access their accumulated earnings on demand. This feature, thus, allowed drivers in the treatment group to reduce the lag between effort and pay by between three and nine days. Whereas standard, time-consistent preferences suggest this shift should have little to no effect on behavior in the absence of sharply binding liquidity constraints (Parsons and Van Wesep 2013), present bias suggests otherwise.

We find that the offer of Instant Pay (intent to treat (ITT)) increases daily labor supply measures by 1.4% (work minutes) and 1.5% (earnings in dollars), both statistically significant at the 5% level. Depending on the measure of take-up used, these results translate to treatment on the treated (TOT) increases in work time of 10.1% (or 21.0%) and in earnings of 10.8% (or 22.8%). Taking the largest of these estimates, we benchmark against a labor supply elasticity estimated by Chen et al. (2019) using trialed wage variation in roughly the same time period and set of drivers. This benchmarking suggests that the effect of Instant Pay among those who activated their cards is roughly equivalent to an 11%

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increase in wages. Several mechanisms could explain this overall rise in labor supply, including sharply binding liquidity constraints and consumption smoothing motives, motivational effects of tying feedback to immediate rewards, and certain forms of present-focused preferences. Whereas each of these mechanisms could be operative for different drivers, we next turn to a set of patterns that implicate the last of these as particularly important.

We present four pieces of evidence pointing toward present bias as an important mechanism underlying the estimated labor supply responses. First, whereas there is ample evidence that many rideshare drivers are liquidity-constrained (Koustas 2018), we do not find significantly larger treatment effects among drivers who reside in less affluent census block groups. Moreover, the magnitude of the effects when benchmarked against wage elasticities imply that, for liquidity constraints alone to account for this effect, drivers would need to act as though they do not have access to credit at less than an 11% weekly interest rate; in conjunction with the fact that all drivers must have a bank account, this explanation seems unlikely for many drivers. Second, we develop a proxy for present bias in the spirit of Kaur et al. (2015) (i.e., whether drivers exhibit a payday effect in labor supply during the pre-experiment period) and show that this indicator strongly predicts take-up; having above-average payday effects in the baseline predicts a 19% increase in take-up (a three percentage point (pp) increase), roughly 2.5 times the size of the relationship with being below the median proxied household income. Third, we examine heterogeneous treatment effects by day of the week and find larger effects on days when the counterfactual pay delay would be larger with a 1.9% ITT on Monday (nine days from payday) and an insignificant −0.4% ITT on Sunday (three days from payday). We use data from another experiment to argue that this pattern does not simply reflect variation by day of the week in labor supply elasticities more generally. Finally, we examine day-of-the-week patterns in utilization of Instant Pay (i.e., cash outs) and find that it does not mimic the one found for labor supply responses. This final result casts doubt on both consumption smoothing motives and standard accounts of present bias over consumption. Instead, these results can be reconciled by present bias over access to cash, consistent with other work showing discounting over cash itself (e.g., Andreoni et al. 2018).

Broadly, our paper contributes to the literature documenting evidence of present bias in the field. Present bias captures the idea that individuals prefer a larger, delayed reward to a smaller, immediate reward when that choice is in the future, but they reverse this preference when the choice is in the present, a phenomenon reflected in hyperbolic and quasi-hyperbolic discounting models.[1] Researchers implicate present bias to explain empirical patterns observed in contexts ranging across consumption (Shapiro 2005), household finance (Angeletos et al. 2001, Meier and Sprenger 2010, Goda et al. 2020, Kuchler and Pagel 2021), and health decisions (DellaVigna and Malmendier 2006, Fang and Wang 2015), among others (DellaVigna 2009). We extend this literature by considering the implications of present bias for a key part of job contract design (pay timing/flexibility).

More specifically, our work relates to a smaller literature on present bias in personnel and labor economics (Cadena and Keys 2022). For example, DellaVigna and Paserman (2005) find that survey-based measures of short-run impatience predict measures of job search and unemployment duration. In perhaps the closest work to our own, Kaur et al. (2015) run a field experiment with data-entry workers and find that they demonstrate strong payday effects and demand dominated (commitment) contracts to solve the self-control problem in effort provision, both results consistent with present bias. Our work extends these by studying an increasingly popular policy that presents another way of addressing the self-control problem; rather than introducing new up-front (conditional) costs to align short-term and long-term preferences, pay flexibility instead brings delayed benefits into the present. This inquiry is especially important in light of evidence that workers demand flexibility in their jobs whether that be over their schedules (Chen et al. 2019) or remote/in-office modality (Hansen et al. 2023). Frakes and Wasserman (2020) show that U.S. patent examiners procrastinate on their tasks (which may account for up to one sixth of the patent backlog) and that this procrastination increases with remote work. As remote work becomes more prevalent—Aksoy et al. (2022) find that job listings in the United States allowing remote work increased threefold from 2019 to 2023—and the external controls of the workplace are relaxed, the role of present bias in labor supply may become more pronounced. Pay flexibility may, thus, play an increasingly important role in labor supply decisions beyond the gig economy.

We also directly contribute to an emerging literature on EWA and early wage access products that allow workers to receive their earnings ahead of payday, usually with a fee. Earned wage access products are offered in partnership with employers and often integrated with existing payroll systems, whereas early wage access products are offered directly to consumers (workers). The market for these products has grown considerably in recent years. For example, the number of transactions processed by earned wage access products grew more than 90% from 2021 to 2022 (Consumer Finance Protection Bureau 2024). Early analysis by Baker and Kumar (2018), studying employee turnover across six U.S. firms partnered with the EWA product PayActiv, estimated a 19% reduction in turnover

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among employees who were active EWA users compared with employees who were enrolled but inactive. This result is consistent with work by Murillo et al. (2023), finding that EWA users at two large Mexican firms were 9%–12% less likely to leave their firms by the next pay cycle compared with nonusers. On the consumption side, De La Rosa and Tully (2022) find that laboratory participants given the ability to access their pay daily in a life simulation task both chose to get paid more frequently and increased their overall spending relative to participants constrained to a weekly pay schedule. Our main contribution is to study the causal effects of the flexible pay provided through an EWA on labor outcomes in a field setting with flexible work (i.e., in which total hours and earnings are adjustable). By benchmarking our treatment effects against findings by Chen et al. (2019) on labor supply responses to randomized wages, we are also able to price our treatment effect estimates. Moreover, our analysis of Instant Pay utilization patterns suggests that drivers respond to the option value of accessing their pay on demand, showing how workers may benefit from EWA even if they do not withdraw their earnings early or often.

Finally, our results relate to a long line of empirical studies of consumption responses to expected and unexpected household income shocks. Mostly aimed at testing the life-cycle permanent income hypothesis, this line starts with Hall (1978) and is summarized in Jappelli and Pistaferri (2010). Recent research using high-quality, account-level data on consumption and borrowing finds surprisingly large (excess) sensitivity of spending to predictable income flows, such as the receipt of a welfare transfer (e.g., Gelman et al. 2014, Baker 2018, Kueng 2018, Olafsson and Pagel 2018, Gelman 2022, Zhang 2022). Much of this work argues that the patterns are best explained by time-inconsistent preferences. More recently, researchers have used financial aggregators to examine the relationship between pay frequency and consumption across both marketing (De La Rosa and Tully 2022) and finance (Baugh and Correia 2022, Baugh and Wang 2022), finding mixed evidence.[2] We extend this line by looking at labor supply rather than spending and borrowing and by looking at pay flexibility rather than pay frequency. That is, we study a policy in which workers can opt into on-demand pay timing rather than the effects of different lengths of fixed schedules set by the employer. As the EWA market continues to grow and more companies begin to offer pay flexibility, the labor supply implications also need to be considered.

2. Background and Experimental Design

2.1. Study Setting

This study took place in partnership with Uber, a popular peer-to-peer ride-sharing platform. Uber drivers use their own or leased vehicles to provide rides to customers at their discretion. The platform imposes minimal restrictions on working hours, allowing for considerable flexibility in labor supply and scheduling.[3] The majority of drivers work part-time; for example, among drivers studied in Chen et al. (2019), the majority work fewer than 12 hours per week, and a substantial fraction who are active in one week are not in the next week. As noted by Hall and Krueger (2018), driving is often complementary to other activities, such as caregiving, employment, and school attendance. Over an eight-month period close to our study period, Chen et al. (2023) note that more than one million individuals worked on UberX, the company's flagship service.

2.2. Instant Pay

Prior to the experiment, Uber paid drivers on a weekly cadence with accumulated earnings from Monday (4 am) to Monday (4 am) deposited on the following Wednesday. Drivers were required to link a checking account to their Uber profile to receive their weekly payments via direct deposit. Starting in March 2016, Uber launched a new feature called Instant Pay to a randomized selection of drivers across several markets (cities) before its national release to all drivers on May 3. This earned wage access product allowed drivers to flexibly cash out their accumulated earnings prior to the usual payday. Specifically, drivers could press a button in the Uber app to initiate a transfer of the entirety of their current earnings balance that would arrive minutes later on a separate debit card. Drivers could complete up to five of these cash outs per day. Importantly, drivers could not cash out a partial amount of their accumulated earnings, and if a driver did not cash out on or prior to Sunday, those accumulated earnings would be sent for direct deposit (and, thus, inaccessible until Wednesday). This Sunday deadline is important for interpreting patterns in cash-out behavior.

During the experimental rollout, drivers in the treatment group who signed up for Instant Pay would receive an Uber debit card from the implementation partner GoBank. The debit card had no fees for cash-outs (transfers from the Uber app to the GoBank account), no withdrawal fees from its network of 42,000 ATMs, and no overdraft/nonsufficient funds/penalty fees. Account maintenance fees (typically $8.95/month) were waived for six months every time the driver received a direct deposit or Instant Pay from the driver's Uber earnings. The underlying GoBank account had limited features compared with other complete checking account products, but it did allow for automated clearing house (ACH) transfers to accounts at other U.S. financial institutions. Meanwhile, Uber's primary rival ride-sharing platform Lyft previously piloted a similar EWA called Express Pay in 2015 in

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conjunction with Stripe. Lyft drivers could use their existing personal debit cards with Express Pay but had to earn at least \$50 before cashing out and pay a \$0.50 fee per cash out (Lyft 2016a).

Uber's EWA continued to pick up usage after the national launch in May, and in August, drivers were able to use most Visa, Mastercard, or Discover debit cards with Instant Pay for a \$0.50 fee per cash out (Koren 2016). By 2017, roughly a year after the launch, it was reported that "hundreds of thousands" of drivers were using Instant Pay and that \$1.3 billion dollars were cashed out in this way (Etherington 2017). By 2019, the head of Uber Money reported that 70% of driver payments were made using Instant Pay (Son 2019). Starting in 2023, GoBank transitioned all existing Uber debit card accounts to GoBank debit card accounts with an \$0.85 fee for Instant Pay cash outs and removed the ability to use ACH transfers (GoBank 2022). Drivers using Visa, Mastercard, and Discover debit cards with Instant Pay are now subject to a \$1.25 fee per cash out, but some high-performing drivers can take advantage of free instant cash outs after each trip using the Uber Pro debit Mastercard and checking account powered by Branch (Uber n.d.).

2.3. Experimental Design and Implementation

Enrollment into the study was done on a rolling basis both within and between cities. Each city had a specific start date on which it began contacting drivers, and roughly 80% of drivers were enrolled into the study on their city's common start date. Upon enrollment, drivers were assigned to treatment or control at roughly an even split. In all cases, treatment entailed access to Instant Pay as well as an email informing the drivers of the program. Online Figure A.1 displays one such example, noting that it is a pilot program and providing a link to the application. The second page of the email included additional details, including a forecast for the time to receive the card if approved (7 to 10 days). Whereas all cities contacted drivers in the treatment group, cities had discretion over whether to email drivers in the control group. Online Figures A.3 and A.4 show an example of the email sent to the control group in the Bay Area market with the tagline note "Coming soon" and a "Keep Me Updated" button instead of an application button. Cities also had discretion over whether to send treated drivers additional emails and SMS messages encouraging program participation, including testimonials from fellow drivers and limited-time bonuses for registering. These messages were generally sent to a random portion of a city's treated drivers within the first two weeks of its common start date, so the volume of communications about Instant Pay varied both across and within cities.

Because drivers were only eligible for the study if they had worked a specific number of hours in the previous two weeks, enrollment was rolling within a city after that date. Each city executed a uniform structured query language query on all drivers who were not yet enrolled in the study. If these drivers newly satisfied an eligibility criterion (based on the number of rides in the previous two weeks exceeding a threshold), they were enrolled, randomized to treatment or control, and contacted accordingly. The randomization is, thus, essentially stratified by date within each city, so in our analysis, we include cohort fixed effects, in which cohort is defined as the date of enrollment interacted with the driver's city.

The experiment ran for two months, beginning in Minneapolis–St. Paul, Minnesota, on March 2, 2016, and gradually expanding across markets. Whereas the initial launch plan intended to randomize across 168 markets in total, Uber pulled the data on April 16 after just 46 markets had begun implementation. Deeming the program an operational success, Instant Pay launched nationally to all drivers earlier than planned on May 3, 2016. As discussed further in Section 3, our data set is composed of the two months pulled by Uber in 2016, spanning February 17 to April 16. Implementation issues in Minneapolis–St. Paul led to imbalances in baseline driving activity across treatment and control groups, so we exclude this market's drivers out of caution.[4] Thus, the first and last implementing cities in our sample are Madison, Wisconsin, and Philadelphia, Pennsylvania, which launched on March 8 and April 12, 2016, respectively.

Online Table A.8 shows the experimental rollout by cities in our sample. The table includes the total number of drivers, modal start date, proportion of drivers enrolled on that date, proportion assigned to treatment, and average number of days drivers in that city are observed after their enrollment date and three weeks thereafter. The last point highlights an issue discussed further in Section 3.3: how to define the post-period. Using three weeks as a data-driven cutoff for when we might expect to see treatment effects implies that only cities implementing at least three weeks prior to April 16 identify our estimates and effectively reduces our identifying variation to the 12 markets that enrolled before this cutoff, beginning with Madison, Wisconsin, on March 8 and ending with Los Angeles, California, on March 18; thus, when we change our definition of post, we expand or contract the set of identifying cities.

3. Data

Our data come from Uber records on drivers enrolled in the study over February to April 2016. We observe a driver's experimental condition and study enrollment date and supplement this information with basic demographic characteristics, such as age, sex, and Uber

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driving tenure. We also include the median household income of the driver's census block group using the 2016 American Community Survey five-year estimates. We then combine information on active minutes worked and payouts associated with trips, Uber debit sign-ups, and Instant Pay cash outs to construct our main measures of labor supply and program take-up.

3.1. Data Construction and Key Variables

Given the staggered rollout of the experiment across city markets, we include fixed effects for the interaction of cohort (city by enrollment date) and calendar date. To do so, we impute the enrolling city for a majority of drivers. We pull possible city candidates from one file on driver demographics and another detailing the cities in which drivers were working at the time. Whereas Uber drivers can only be actively registered with one market, drivers can work anywhere within the same state and change with which local market they are registered. By comparing each candidate city's modal experiment start date with a driver's enrollment date, we assign a city to a driver if a candidate city's common experiment start date is prior to and the closest to the driver's start date. When we cannot assign a city based on this criteria, we place drivers in a catch-all "synthetic city" to avoid dropping drivers. There are also a handful of unnamed cities that cannot be matched to cities in the launch plan, and we do not modify assignments for drivers linked with these unknown cities. Fewer than 1% of drivers in our final sample are assigned either the synthetic city or an unnamed city, and dropping these drivers has no meaningful effect on our estimates.

Our first labor supply measure is minutes worked per driving session aggregated over a calendar day, for which driving sessions are blocks of time in which drivers are actively completing UberX, Uber Pool, and Uber Eats trips. Sessions may include breaks in which we do not observe an active trip, and we use dormant periods exceeding two hours to delineate driving sessions. Minutes worked per session is then defined as the length of the entire session (i.e., the difference between the drop-off time of the last ride in a session and the pickup time of the first ride of the session). Defining session earnings is more straightforward and calculated by summing net earnings over the corresponding trips for which net earnings is the trip's ride fare less Uber's commission or a fixed \$3.00 per Uber Eats trip.

To construct our daily-level outcomes, we analogously aggregate session minutes and session earnings within a specific date for our daily session minutes and earnings outcomes. The daily session count is the number of sessions on a given calendar date. Note that sessions are assigned to the day in which they began such that a driver who nets \$50 from 8 am to 10 am on March 3 and \$100 from 10 pm into 2 am of the following date has two driving sessions, 360 minutes worked, and \$150 assigned to March 3.

As discussed by Chen et al. (2019), there are various ways to define labor supply for Uber drivers, and each carries its own drawbacks. We believe that our method strikes a reasonable balance to estimate the time spent working that is not captured by the trip time stamps, such as waiting for ride requests. Similarly, defining working time at the session level avoids overestimating labor supplied when drivers can work multiple trips simultaneously. For example, drivers may generate several Uber Pool trip records with nested or overlapping start and end times, and simply summing up trip length would double count minutes worked in these cases. Note that the pulled trips data has coverage issues after April 16: there are no trips recorded between April 17 and 25, and the data remain sparse through the rest of the month. Because of the missing dates, we cut off the sample on April 16. However, in Online Appendix B, we discuss a way to extend the sample for the final two weeks of the experiment by using data from Chen et al. (2019).

For treated drivers, we observe the dates they reached various stages of the Uber debit funnel: submitting an application to GoBank and, if approved, registering for an account, activating the account, and performing a cash out by pushing a button in the driver app to transfer all accumulated earnings to the GoBank account. Not all drivers interested in taking advantage of the newly offered payment flexibility were able to activate a card. Around 4.1% of treated drivers who applied were declined, and GoBank surveys report a mix of reasons why people fell off at different stages, such as technical issues completing the sign-up process, confusion around fees, and diminished interest. To examine take-up over time, we create time-varying measures of take-up for the earliest date drivers reach each of these phases; however, for our take-up analysis and for estimating the treatment on the treated, we use time-invariant measure of take-up reflecting whether a driver ever reaches that stage on or before April 16.

We also count the daily number of Instant Pay cash outs and total amount withdrawn for each driver. These cash outs are transfers out of Uber earnings as Instant Pay deposits to the driver's GoBank account. Drivers could access their Uber earnings before they received their Uber debit cards if they initiated an ACH transfer from GoBank to another financial institution. Such transactions were subject to standard processing wait times and limits on the number of bank transfers within a given period and, thus, offered limited payment flexibility compared with the full Instant Pay product.

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3.2. Balance Analysis and Summary Statistics

Our final data set is a balanced panel of 222,162 drivers spanning February 17 to April 16, 2016. Online Table A.1 provides driver-level baseline (pre-experiment) summary statistics and balance by treatment condition. *T*-tests of the four demographic variables (sex, age, median household income of driver's home census block group, and driver's previous Uber experience) fail to reject the null hypotheses of equality between the treatment and control groups.[5] The same is true for the measures of labor supply in the two weeks prior to the experiment's first planned enrollment date (i.e., February 17 to March 1). Finally, we report an *F*-test for joint significance, and this also fails to reject the null ($F = 1.02$). This specification includes cohort fixed effects to match our regression analysis, but omitting the fixed effects produces a similar *F*-statistic (1.08).

The table shows that 16% of drivers are female, the average age is 43, and the typical driver lives in a census block group with a median income of $65,000 dollars. Drivers in our sample average one session of work every two days for 111 minutes and $38 per day. This activity aggregates to average totals of roughly eight shifts, 1,660 minutes, six days worked, and $574 in this baseline period.

3.3. Take-Up and Posttreatment

Because converting an emailed offer of Instant Pay into actual access to earnings requires a number of steps, a key issue is how to define when the posttreatment period starts. Defining post too early may attenuate treatment effects by identifying off days in which the driver lacks the ability to use the service. This is because there are access delays because of at least three margins: (a) the driver needs to apply (requiring first reading the email); (b) GoBank has to review the application to decide if the driver is approved; (c) after approval, there are delays in receiving the debit card. On the other hand, defining the post period too late risks inflating standard errors as there are fewer days identifying the treatment effect. We, thus, face a bias–variance tradeoff in this decision.[6] Because our postimplementation observation window is short (roughly 32 days on average), this issue is of first order importance.[7]

To take a data-driven approach to defining the post period, in Figure 1, we look at cumulative take-up rates by the number of days since a driver enrolled in the experiment. This figure suggests that take-up, as proxied by application and activation, appears to steadily increase between 10 and 28 days after enrollment. There is no clear cutoff point across all four measures of take-up, but 21 days appears to somewhat balance the concerns, so we make this our primary measure of post. In Section 5.3.1, we show robustness to varying this definition from 0 to 30 days.

Panel A of Online Table A.2 shows take-up rates by treatment group among the 55,841 drivers who have at least one observation 21 days post their experiment enrollment. The take-up rates in this group are relevant for our primary estimation sample and for comparing the intent-to-treat and treatment-on-the-treated estimates. Take-up measured by application is 13%, by registration is 12%, by activation is 6%, and by ever making a cash out before the study ending is 4%. The table also shows that there is only one-sided noncompliance as the take-up rates are zero in the control group.

Finally, panel B of Online Table A.2 telegraphs our regression estimates by showing simple *t*-tests of differences in the outcomes using just posttreatment observations, capturing the ITT effects. Using this driver-day–level sample, we see that work time increases by 1.89 minutes (roughly 1.4% of the control group mean), earnings by $0.81 (or 1.6%), and session count by less than 0.01 (less than 1.6%). In Section 5, we estimate these treatment effects using the full panel structure.

4. Empirical Strategy

We first examine the impact of the randomized offer of Instant Pay on driver outcomes, that is, the ITT of Instant Pay. We use the full two months of our panel, and thus, because implementation is staggered, the number of pre-enrollment versus postenrollment observations varies within and between cities. For example, implementation began in Houston on March 30, so we have six weeks of pre-enrollment data and roughly two weeks of postenrollment data for that city. However, 20% of Houston drivers were enrolled after that date and so have fewer days postenrollment. As discussed in Section 2, we include fixed effects for the cohort of enrollment (i.e., each distinct randomization) interacted with date fixed effects to isolate treatment versus control differences in the posttreatment period, accounting for dynamics common to both groups within a cohort over time. Preperiod observations are included for their potential to increase precision and not for causal identification (as treatment is randomized), but we show in Section 5.3.3 that our estimates are quite similar if we limit to posttreatment observations and omit fixed effects. We define the posttreatment period as 21 days after enrollment to account for delayed sign-up and mailing of debit cards and show robustness to this choice in Section 5.3.1.

Specifically, we estimate the following driver-day–level regression specification:

$$y_{izj} = \beta_1 * Treatment_{iz} * Post_{izj} + \eta_{zj} + \epsilon_{izj}, \qquad (1)$$

where y_{izj} is a measure of daily labor supply (e.g., work minutes or earnings) for driver i working in city z on date j, regressed on an indicator for whether the driver is in the treatment group ($Treatment_{iz}$) interacted with a

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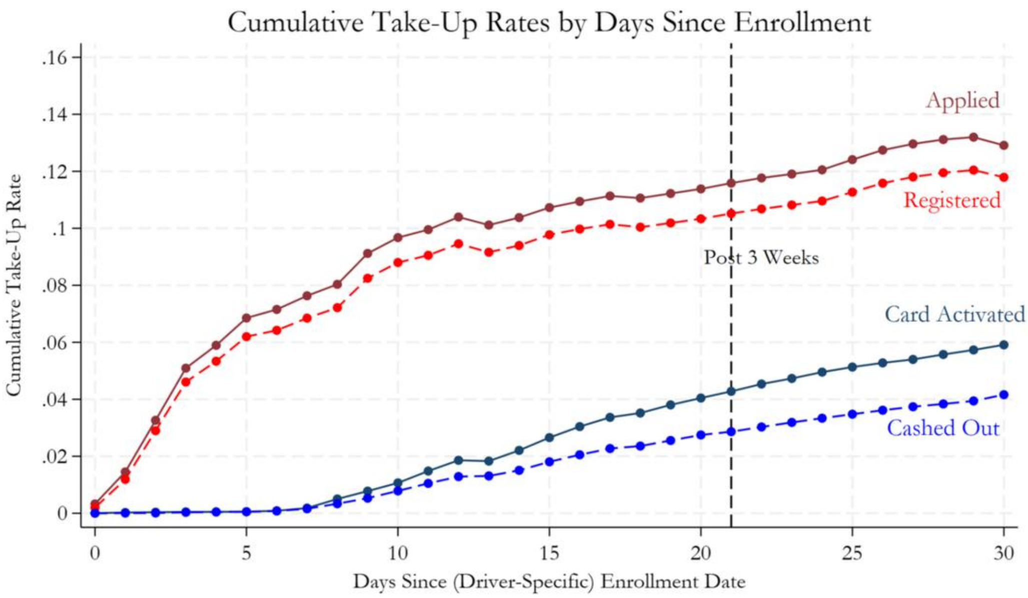
Figure 1. (Color online) Take-up Rates by Days Since Enrollment (Launch Email Date)



Notes. This figure's sample is composed of drivers in the treatment group and includes one observation per driver-day. Each point corresponds to the mean on the day since the (driver-specific) enrollment date. For example, if one driver was enrolled into the study on March 8 and another started on March 10, day 1 would correspond to March 9 and 11, respectively, for those drivers. Note that, because this is a cumulative measure and the postenrollment data windows vary by cohort/cities, the denominator can change throughout (e.g., explaining the downward dip between days 12 and 13).

time-varying indicator variable that takes on the value one three weeks after the driver's study enrollment date ($Post_{izj}$) as well as a set of cohort-by-date fixed effects (η_{zj}). We cluster standard errors at the unit of randomization (driver). The key coefficient from Equation (1), the reduced-form parameter β_1, captures the effect of being offered Instant Pay.

We examine outcomes in terms of levels as well as with specifications that allow interpretation in percentages in order to facilitate comparisons to other labor supply elasticities. For the latter, because there are many potential zero-valued outcome observations, we use Poisson regression because of its favorable econometric properties (Wooldridge 2010, Chen and Roth 2024).

As discussed in Section 3.3, take-up is somewhat mechanically depressed because of the short observation window and implementation challenges. Because we're interested in the labor response of drivers actually capable of taking advantage of the greater pay flexibility, we also present TOT estimates, scaling up the ITT to account for the fact that not all drivers took up the treatment before the end of our observation window. That is, we estimate instrumental variable specifications for which the second stage is

$$y_{izj} = \beta_1 * Takeup_{iz} * Post_{izj} + \eta_{zj} + \epsilon_{izj}, \qquad (2)$$

where $Takeup_{iz}$ is an indicator that takes value one if a driver took up Instant Pay at any point prior to April 16. Of course, take-up is not random and may be correlated with unobservables, so we instrument $Takeup_{iz} * Post_{izj}$ with the variable $Treatment_{iz} * Post_{izj}$, which is uncorrelated with the error because of randomization. As with the ITT, $Post_{izj}$ is defined as 21 days after the driver-specific enrollment date, allowing a simple comparison of the ITT and TOT (i.e., the latter point estimate is equal to the ITT divided by the take-up rate).

We show TOT estimates in which *Takeup* is defined as either (a) applying for the card or (b) activating the card after receiving it in the mail. Each of these definitions has trade-offs, which we discuss further in Section 5.2. The key identification assumption is that the assignment of treatment only affects labor supply through the channel of take-up as measured by either the application in case (a) or the activation of the card in case (b). Again, our preferred estimates use Poisson regressions, and we adopt a control function approach and use the delta method to calculate standard errors as suggested by Lin and Wooldridge (2019). We estimate the first stage via ordinary least squares (OLS) regression and add residuals from the first stage as a control in the second stage Poisson regression. In Online Table A.4 we reestimate this specification using just the posttreatment observations (and omitting fixed effects) and obtain qualitatively similar estimates.

5. Main Results

This section describes the impacts of Instant Pay on driver labor supply. We start by reporting the ITT

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estimates. Then, we turn to the TOT, and we show results using two possible measures of take-up: ever applying for Instant Pay and ever activating an Uber debit card by the end of our observation window. We show all results using Poisson and OLS regressions.

5.1. ITT

Table 1, panel A, reports the ITT estimates of Equation (1). Columns (1)–(4) report our preferred estimates using Poisson regression, and columns (5)–(8) report OLS specifications. Our first measure of labor supply is the number of session minutes worked over a day. Column (1) shows that the offer of Instant Pay increased daily labor supply by 1.4% in terms of minutes worked, and column (2) reflects a similar 1.5% increase in earnings. Note that work time and earnings are not constrained to move together because of drivers' flexible choice of when to supply labor and Uber's dynamic pricing algorithm. For example, if drivers already worked at times of day with the highest earnings opportunities, then the marginal product of labor would be diminishing. However, we do see similar increases in both outcomes that are statistically significant at the 5% level, suggesting that drivers may have increased work time without reducing their average productivity. In columns (3) and (4), we disaggregate the increase in work time into two margins: whether the driver works at all on a given day (column (3)) and the total time worked on a day conditional on working at all (column (4)). Of course, because the decision to work is downstream of treatment, column (4), which conditions on this variable, should be interpreted with caution as it may suffer from posttreatment bias (Montgomery et al. 2018). Column (3) shows that the probability of working any sessions increases by 0.6%, and column (4) shows that work time conditional on working increases by 0.8% with only the latter being significant (at the 5% level).

Columns (5)–(8) report results of OLS regressions in levels. Minutes worked increased by 1.91 minutes, and earnings increased by \$0.76. The daily probability of working increased by 0.3 pp, and work minutes conditional on working increased by 2.38 minutes. Dividing these point estimates by the preperiod mean of the dependent variable in the control group implies that these outcomes increased by 1.6%, 1.8%, 0.7%, and 0.8%, respectively. These results are all quite similar to the Poisson regression results and demonstrate that the ITT estimates are robust to the estimation method. Online Table A.7 reports estimates using a $\log(1 + Y)$ transformation that produces similar (slightly larger) estimates.

Table 1. Main Results

	Poisson				OLS (levels)			
	Minutes (1)	Dollars (2)	Work (3)	Minutes | Work (4)	Minutes (5)	Dollars (6)	Work (7)	Minutes | Work (8)
				Panel A: ITT				
Treat × Post	0.014**	0.015**	0.006	0.008**	1.905**	0.757**	0.003	2.375**
	(0.006)	(0.006)	(0.004)	(0.004)	(0.858)	(0.323)	(0.002)	(1.084)
N	13,242,030	13,241,824	13,242,030	5,795,535	13,329,180	13,329,180	13,329,180	5,795,535
DepVarMean	121.752	42.082	0.435	280.039	120.885	41.781	0.432	280.039
				Panel B: IV (TOT, Take-up = Applied)				
Applied × Post	0.101**	0.108**	0.040	0.060**	14.159**	5.622**	0.019	16.893**
	(0.045)	(0.046)	(0.032)	(0.027)	(6.379)	(2.400)	(0.015)	(7.715)
N	13,242,030	13,241,824	13,242,030	5,795,535	13,329,180	13,329,180	13,329,180	5,795,535
DepVarMean	121.752	42.082	0.435	280.039	120.885	41.781	0.432	280.039
				Panel C: IV (TOT, Take-up = Activated)				
Activated × Post	0.210**	0.228**	0.078	0.130**	30.254**	12.013**	0.042	32.947**
	(0.097)	(0.098)	(0.067)	(0.057)	(13.627)	(5.127)	(0.032)	(15.059)
N	13,242,030	13,241,824	13,242,030	5,795,535	13,329,180	13,329,180	13,329,180	5,795,535
DepVarMean	121.752	42.082	0.435	280.039	120.885	41.781	0.432	280.039

Notes. Panel A reports ITT estimates from regressions with fixed effects for the interactions of randomization cohort (city-by-enrollment-date) and date. The "Post" period is defined as beginning three weeks (21 days) after the (driver-specific) enrollment date. Columns (1)–(4) report Poisson regressions, whereas columns (5)–(8) report OLS of the untransformed outcomes. Minutes and Dollars are the daily totals for minutes worked and dollars earned. Work is an indicator for having driven any sessions on a particular day. Minutes | Work represents regressions of daily minutes worked restricted to days when drivers worked any shifts. Panel B reports instrumental variable estimates using the randomized assignment of treatment as an instrument for take-up (defined as the driver have ever applied for a card prior to April 16; mean = 13%). Panel C is similar but uses having ever activated a card as the definition of take-up (mean = 6%). Columns (1)–(4) of panels B and C use a control function approach in which the first stage is estimated via OLS (*reghdfe*), residuals are included as a control in the second stage Poisson regression (*ppmlhdfe*), and the delta method is used to compute standard errors (using *margins, eydx*). DepVarMean is the mean in the control group in the preperiod. Standard errors are clustered by driver.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

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5.2. TOT

Panels B and C report the TOT estimates of Equation (2), in which the randomized offer of treatment instruments for take-up. Panel B focuses on the loosest measure of take-up, which is whether the driver ever submitted an application. In panel C, the measure of take-up is whether the driver ever activated the driver's Uber debit card, which required first applying for the card and registering for GoBank online. As shown in Online Table A.2, roughly 13% of drivers in the sample ever applied and roughly 6% of drivers activated an Uber debit card before April 16. Among drivers who ever activated a card, it took roughly 7.1 days on average for drivers to apply for the card after receiving the first enrollment email and roughly 18.2 days to activate their card.

It is not obvious which TOT measure is most relevant a priori. On one hand, cashing out seems a natural candidate as it reflects the most direct measure of using the EWA. After all, drivers who do not cash out have not had an actual change in their pay flexibility. However, it is plausible that drivers shift their labor supply after activating their card as it is the presence of the card that changes the ability to access additional earnings in the event of a cash need that may not yet have arisen before the end of our observation window. For this reason, we prefer activation over cash out as a measure of access. However, even activation comes well after drivers have taken a formidable step to apply for the program, and drivers may adjust their behavior in anticipation before receiving the card.

Columns (1)–(4) of panel B, Table 1, using *Applied* as a measure of take-up, finds TOT effects of daily work time increasing by 10.1%, earnings by 10.8%, the daily probability of working any sessions by 4%, and daily work time conditional on working by 6.0%. Except for the effect on the probability of working, these TOT estimates are statistically significant at the 5% level. In terms of levels estimated via OLS (columns (5)–(8)) and percentages relative to the control group mean, these are increases in worktime of 14.2 minutes (12%), earnings of $5.62 (13%), probability of working of 1.9 pp (4%), and worktime on days drivers work of 16.9 minutes (6%). Using card *Activation* as the measure of take-up, panel C shows work time increasing by 21.0%, earnings by 22.8%, probability of working by 7.8%, and worktime on days drivers work by 13.0%, and only the effect on the probability of working is insignificant. The level increases are 30.3 minutes (25%), $12.01 (29%), 4.2 pp (10%), and 32.9 minutes (12%), respectively.

5.3. Robustness

5.3.1. *Post* Window Definition. A natural question is how to define the *Post* period. As noted in Section 4, if we define *Post* too early for drivers to receive their cards, then we potentially attenuate the treatment effect. By contrast, if we define *Post* too late, then we both reduce power by limiting the observation window and may mischaracterize the total treatment effect if there is dynamic heterogeneity. In our primary analysis, we settled on 21 days after enrollment to define *Post*; however, because this choice was not preregistered, one may be concerned about specification search. Figure 2 repeats the specification shown in Table 1, panel A, column (1), for 31 possible definitions of *Post*, ranging from 0 to 30 days. We see that the effect remains relatively stable and significant at the 5% level across most of the range of values (and significant at the 10% level across the entire range).

5.3.2. TOT Estimation. Another degree of freedom is how to estimate the TOT given the staggered rollout and endogenous timing of take-up by drivers. We choose to define "ever take-up" as a time-invariant property of a driver and to instrument it with the similarly time-invariant treatment indicator. This approach has the desirable property of preserving the usual cross-sectional logic that the TOT is simply the ITT divided by the take-up rate. By contrast, another approach found in the literature (e.g., Atkin et al. 2017) is to instead construct a time-varying measure of take-up that turns on when the driver chooses to apply or activate the card and to then simply instrument this with the binary treatment indicator. Because this scaling mixes both the take-up rate and the endogenous timing of take-up, it does not facilitate as straightforward a comparison with the ITT. Nonetheless, for completeness, we estimate this specification in Online Table A.3. Comparing the primary Poisson IV specifications to Table 1, we see that this approach produces larger coefficients on minutes (13.7% versus 10.1%), dollars (13.3% versus 10.8%), and minutes conditional on working (11.8% versus 6%) but crosses the significance threshold, whereas the effect on the probability of work is smaller and remains insignificant (1.8% versus 4.0%). When take-up is defined as activation, the differences are even larger across minutes (50.7% versus 21.0%), earnings (48.3% versus 22.8%), and minutes conditional on working (44.2% versus 13%) and slightly smaller (6.4% versus 7.8%) for the probability of work, but again, these suffer from a lack of precision.

5.3.3. Use of Pretreatment Observations and Fixed Effects. We next investigate whether results are similar when omitting pretreatment observations. Our primary specification uses the balanced panel spanning pretreatment to posttreatment days to potentially increase the precision of our estimates; however, this choice requires including cohort fixed effects to ensure the staggered rollout does not introduce bias. The complication this poses is that estimating the TOT in our preferred Poisson specification requires a control function

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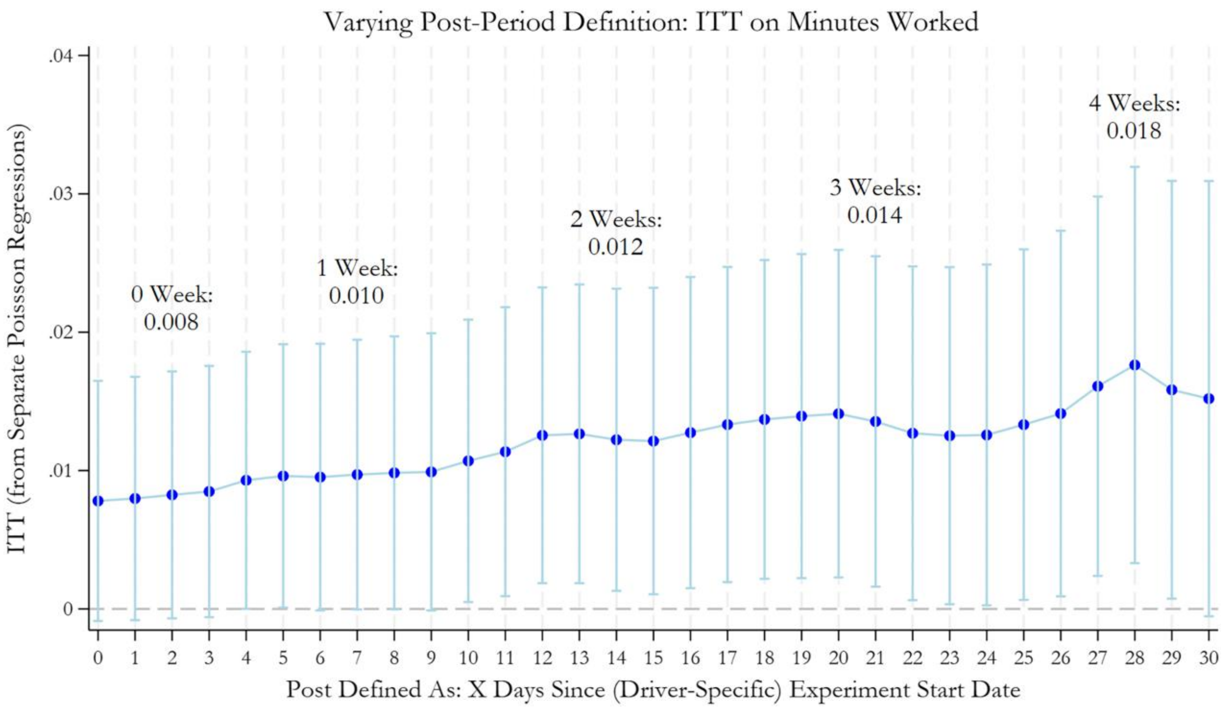
Figure 2. (Color online) Robustness to *Post* Window Definition



Notes. Each dot corresponds to a coefficient on the interaction term (*Treat* × *Post*) from a separate Poisson regression for each "Post" definition, ranging from 0 to 30 days. The bars correspond to the 95% confidence interval (±1.96*SE) on that point estimate. The Poisson regressions include fixed effects for the interactions of randomization cohort and date. Standard errors are clustered by driver.

approach and using the delta method to compute standard errors. By contrast, if we only use the posttreatment observations, we could more safely drop the fixed effects and use a two-step generalized method of moments estimator to calculate the correct clustered standard errors. In Online Table A.5, we reproduce Table 1 using just the observations at least 21 days after driver enrollment and omitting any fixed effects. Panel A shows that this approach produces quite similar ITT point estimates (e.g., 1.3% versus 1.4% for minutes in column (1)). Columns (1)–(4) of panels B and C are our main interest, and we again see quite similar point estimates (e.g., 9.6% versus 10.1% for minutes in panel B and 19.5% versus 21.0% in panel C) and standard errors.

We also examine whether our results are sensitive to omitting date fixed effects from the interaction with cohort fixed effects. Using the full balanced panel, we limit the set of controls to cohort fixed effects and a single binary indicator for the three-week-post period (to isolate treatment versus control differences after treatment begins). Online Table A.5 shows that this alternative specification produces extremely similar point estimates and standard errors to those in our main table.

5.3.4. Extending the Sample to Examine Dynamics. Finally, we consider how treatment effects evolve over time as a cohort gets further from the enrollment date. If there are novelty effects, learning, or other types of dynamics, then it is possible that the treatment effects estimated over our fairly short window overstate or understate the effects that may be observed over a longer horizon. To investigate, we trace out the evolution of treatment effects estimated over each week past enrollment. The main challenge to this exercise is the limited posttreatment observation window. To partly address this shortcoming, in Online Appendix B, we explain how we can fill in the final two weeks of the experiment using overlapping driver-hour–level data from Chen et al. (2019), albeit with some necessary adjustments. This additional data allows us to track some Uber drivers as far out as seven weeks after entering the experiment.

To explore treatment effects over time, we construct an event study-like figure that modifies the regression in Table 1, panel A, column (1) by interacting the *Treatment* indicator with indicators for each week (defined as seven days) prior to and after enrollment. Because of the staggered rollout, we have at most eight complete weeks prior to and seven complete weeks after entering the study. We estimate all weekly treatment effects relative to one week prior to enrollment. The top panel of Online Figure B.1 shows that, starting in week 0 when enrollment occurs, the ITT estimate steadily grows over time and first achieves statistical significance at the 5% level in week 3. It continues to rise and remains

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significant through week 5. The final two weeks produce point estimates similar to week 5, but because there are fewer observed drivers this far out from enrollment (i.e., just the early enrolling cities), the confidence interval widens.

One challenge to the interpretation of this figure is that the staggered rollout of the experiment means that any comparison between weeks may conflate actual dynamics in treatment effects with heterogeneity in treatment effects across cities/cohorts as the composition of which drivers/cities contribute to these estimates varies over time. To address this issue, we also estimate the same specification, limiting to a set of drivers who started early in the study and entered on roughly the same date. Specifically, we focus on drivers enrolled on their city's modal experiment start date in the four earliest cities: Seattle (March 8), San Diego (March 8), Madison (March 8), and Boston (March 10). Whereas noisier, we again see that the treatment effect does not wane toward the end of this window. Together these results suggest that our treatment effect magnitudes are not simply driven by the initial novelty of the program insofar as such novelty wears off within these first seven weeks.

5.4. Magnitude

One potential complexity in interpreting the magnitude of our results is that we observe only on-platform labor supply, and most Uber drivers also do non-Uber work. This limitation may complicate the interpretation of our measured elasticities; is Instant Pay making working on Uber more attractive in absolute terms or only relative to other work? Perhaps most importantly, many Uber drivers also work on non-Uber gig platforms, and substitution away from these may represent relatively modest changes in the absolute attractiveness of work on Uber.[8]

To provide context for our results that helps address this complexity, we first consider how the observed labor supply responses compare with previously estimated wage elasticities among Uber drivers (Chen et al. 2019). Labor responses to experimentally randomized increases in Uber wages share the same interpretational issues we discuss above, but we can both ask if responses to Uber wages seem nonnormatively high and also translate Instant Pay labor-supply responses to equivalent increases in Uber wage rates. Chen et al. (2019) find a median driver-wage labor supply elasticity of 1.92 (25th percentile: 1.81; 75th percentile: 2.01). As noted there, these elasticities are somewhat high relative to estimates from the labor economics literature, but they note that the bulk of Uber drivers in their sample work roughly 10 hours per week and, thus, have more scope for adjustment than other classes of workers.

To translate our estimates to a wage elasticity, it is not a priori obvious whether to use the ITT or TOT. Whereas wage shifts have perfect compliance as they apply to all drivers, the programmatic costs of Instant Pay (e.g., subsidizing transaction fees) are only borne by the drivers who use it. This consideration pushes in the direction of benchmarking relative to the TOT for considering cost-effectiveness. Similarly, for thinking about the behavioral mechanism, the TOT may be closer to the right individual-level estimate as the lower ITT is mechanically driven to zero by low take-up during the short pilot period. On the other hand, there are reasons the TOT may not extrapolate to the majority of drivers who eventually take up Instant Pay. Namely, we know they differ on observable characteristics. Table 2 predicts the probability of treated drivers reaching various take-up stages and shows that drivers who ever applied and activated their cards were more likely to be female (2.9 pp and 1.1 pp) and younger (−3.6 pp and −2.2 pp for being older than the median age of 42). For predicting application, being above the median income was also a significant negative predictor of take-up (e.g., −1.1 pp for having proxied income exceed $58,750). What may be more relevant are unobservable differences from those who take up later. For example, early adopters may have had the highest valuation for Instant Pay (and, thus, possibly the largest local average treatment effects); however, it is also plausible that early adopters are the most attentive to emails or notifications. The correlation between attentiveness and valuation of Instant Pay is unclear, meaning it is also possible that our TOT may underestimate the TOT for individuals who eventually adopt. As a result, both ITT and TOT benchmarks may be useful.

If we use 1.92 as the wage elasticity, this elasticity implies that the ITT estimate of a 1.4% (work minutes) labor supply response to the offer of Instant Pay is roughly equivalent to raising wages for all drivers by 0.7% (0.014/1.92). Similarly, the TOT using application (panel B of Table 1) of 10.1% is roughly equivalent to raising wages by 5.2%, whereas the TOT using activation of the card (panel C of Table 1) of 21.0% is equivalent to raising wages by 10.9%. Whereas we cannot observe outcomes after the full rollout of Instant Pay or estimate how much of the labor supply response came from substitution away from other forms of work, all of these wage benchmarks suggest that the EWA makes work on Uber considerably more attractive to Uber drivers.

6. Mechanism

What mechanism can explain the observed labor supply responses to flexible pay? Instant Pay made the effort costs and financial benefits of work roughly coincident, reducing the lag between them by several days.

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Table 2. Take-up Analysis

	Ever applied		Ever registered		Ever activated card		Ever utilized	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
High payday effects	0.0252***	0.0253***	0.0245***	0.0246***	0.0142***	0.0144***	0.0127***	0.0129***
	(0.0030)	(0.0031)	(0.0029)	(0.0030)	(0.0021)	(0.0022)	(0.0018)	(0.0019)
Female		0.0285***		0.0274***		0.0114***		0.0072***
		(0.0045)		(0.0044)		(0.0032)		(0.0027)
Older than median driver age		−0.0357***		−0.0308***		−0.0218***		−0.0213***
		(0.0030)		(0.0029)		(0.0021)		(0.0018)
Above median driver median household income (home block group)		−0.0108***		−0.0098***		−0.0008		−0.0015
		(0.0031)		(0.0030)		(0.0022)		(0.0018)
Above median Uber experience (months)		0.0028		0.0031		0.0033		0.0053***
		(0.0031)		(0.0030)		(0.0022)		(0.0018)
Full-time driver		−0.0174***		−0.0177***		−0.0124***		−0.0090***
		(0.0044)		(0.0042)		(0.0030)		(0.0025)
N	55,833	53,182	55,833	53,182	55,833	53,182	55,833	53,182
DepVarMean	0.132	0.134	0.121	0.123	0.061	0.061	0.043	0.043

Notes. This table reports estimates from OLS regressions predicting various take-up measures among treated drivers who were at least three weeks post their enrollment date on April 16, 2016. High payday effects is a proxy for driver's impatience in the baseline preperiod adapted from the measure used by Kaur et al. (2015). This variable equals one if a driver's difference in mean daily minutes on Sundays and Mondays in the baseline divided by the mean daily minutes over the entire baseline period is above the sample mean. We also include indicators for median splits on age (42 years), proxied income ($58,750), and Uber driving experience (16 months). Full-time driver corresponds to the 12.2% of treated drivers who average working at least 35 hours per week in the common baseline period of February 17 to March 2. The take-up regressions include fixed effects for the randomization cohort (city-by-enrollment-date), and we report robust standard errors.
*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

As such, a leading hypothesis is that the response relates to time preferences. However, several potential mechanisms can explain the overall increase in labor supply including—but not necessarily limited to—liquidity constraints and consumption-smoothing motives, motivational accounts related to the immediacy of rewards, mental (and actual) accounting, and temporal discounting. Ultimately each of these mechanisms may be more or less operative for different drivers; however, in this section, we present a set of findings that implicate present bias as particularly important for understanding the average response.

First, we consider consumption-smoothing motives by revisiting the benchmarking exercise of the previous section and estimating heterogeneous treatment effects. The TOT (using card activation) is equivalent to the response expected for an 11% increase in wages. Because the product effectively reduces the average delay between effort and pay by six days (weighting each day of the week equally), this result implies a six-day discount factor of 0.9. Thus, an exponential discounting model without binding liquidity constraints implies an annual discount factor of 0.00181 ($= 0.90143^{365/6}$) or an annual discount rate of 55,064%. Either sharply binding liquidity constraints or present bias could help rationalize this large response without requiring a large long-term discount rate. However, on the former, recall that all drivers had bank accounts as it was required to receive a direct deposit. Thus, whereas many drivers are quite liquidity constrained (Koustas 2018), it is plausible that they have access to credit at a rate cheaper than 11% for a six-day repayment period. Nonetheless, we examine whether treatment effects are larger for drivers more likely to be liquidity constrained. Whereas we cannot observe liquidity directly, we use a proxy for their household income (the census block group median household income of their home address). It is worth noting, however, that even if we did observe liquid assets, it would still be challenging to disentangle time preferences from liquidity as the former can drive the latter, a point well made by Gelman (2022).[9] More broadly, Figure 3 repeats the ITT Poisson specification reported in column (1), panel A of Table 1 for the full sample and various subsamples. In particular, we find little evidence for heterogeneity across all four outcomes by the sample splits. These include sex (male/female) and median splits on age (42 years), proxied income ($58,750), experience (16 months), baseline Uber usage (0.4 average sessions per day in the common preperiod from February 17 to March 2), and full-time work status on Uber (i.e., splitting on the 12% of drivers who average more than 35 hours per week in the common preperiod). Of particular note is the lack of heterogeneity by the median income of the drivers' home block group. Notwithstanding the caveat that observed liquidity constraints themselves could be driven by time preferences, we expect a larger response among lower income drivers if liquidity constraints alone drove the labor supply response, but we do not see this result directionally or statistically. More broadly, we find no reliable differences across any

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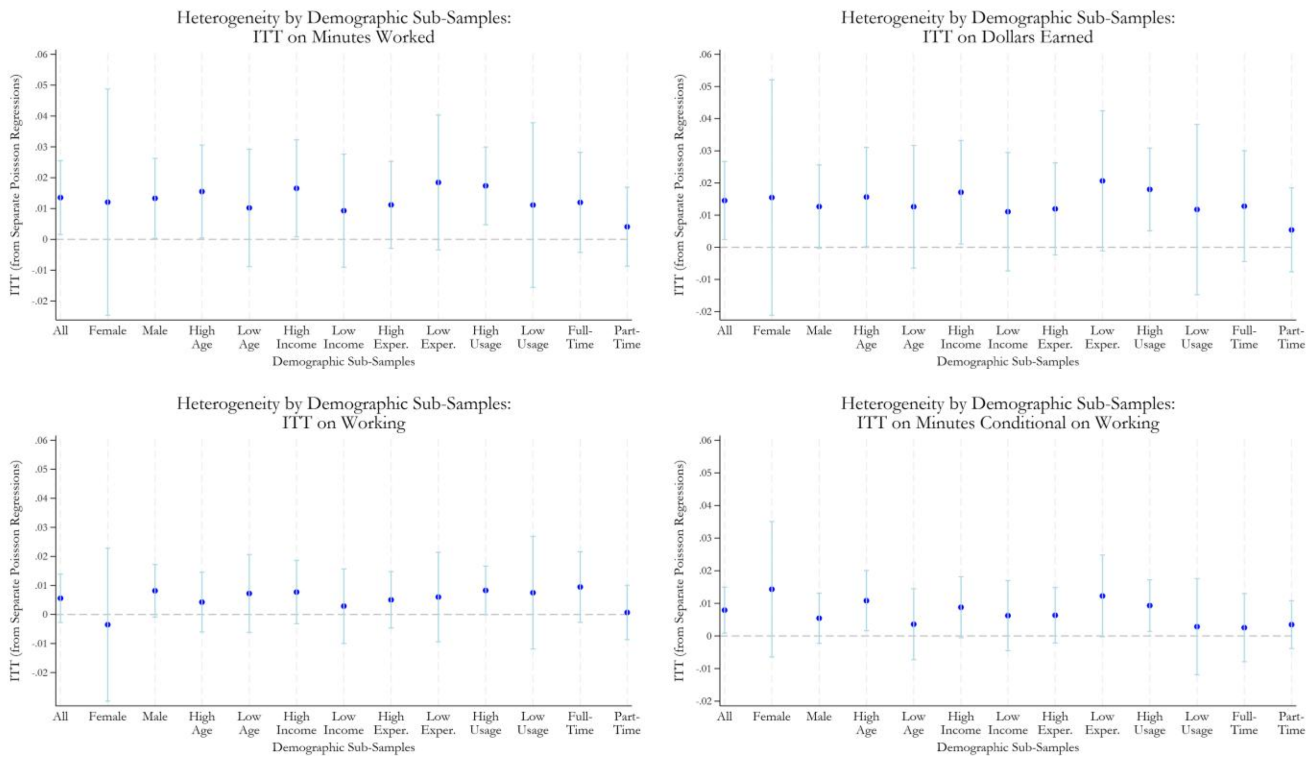
Figure 3. (Color online) Heterogeneity by Demographic Subsamples (ITT)



Notes. Each dot corresponds to a coefficient on the interaction term (*Treat* × *Post*) from a separate Poisson regression for each subgroup subsample. The bars correspond to the 95% confidence interval (±1.96*SE) on that point estimate. Regressions include fixed effects for the interactions of randomization cohort and calendar date. Standard errors are clustered by driver.

of the heterogeneity cuts. Whereas the lack of heterogeneity by income could be partially explained by attenuation bias from using a noisy proxy, these results are at least suggestive evidence against an account that only includes liquidity constraints.

Second, we revisit the take-up analysis with a focus on a proxy for present bias constructed from behavior in the common pre-experiment baseline window (February 17 to March 2). This proxy is similar to the measure of "high payday effects" used by Kaur et al. (2015) to predict take-up of dominated pay contracts among piece-rate data-entry workers. For each driver, we take the difference in mean minutes worked on Sundays (earnings paid three days later) and Mondays (earnings paid nine days later) in the baseline period and divide it by the driver's daily mean over the entire baseline period. We then define an indicator for high payday effects that equals one if a driver's baseline payday effect was above the sample mean. Unlike in Kaur et al. (2015), Uber drivers' paydays in the baseline are not randomized; however, we believe this scaled difference in means captures the spirit of their payday effects measure: time-inconsistent workers are more productive (i.e., supply more labor) the shorter the time interval between work and the receipt of payment for that work. Table 2 shows that having above-average payday effects is associated with increases of 2.53 pp for ever applying, 2.46 pp for ever registering, 1.44 pp for ever activating a card, and 1.29 pp for ever utilizing Instant Pay. In percentage terms, these are increases of 18.9%, 20.0%, 23.6%, and 30.0%, respectively, and all estimates are statistically significant at the 1% level. Thus, the more likely a driver was to supply more labor the closer they were to being paid the higher the driver's likelihood of taking up Instant Pay. The size of these correlations is roughly equivalent to the associations with being female and greater than twice the effect of being above the sample median on the proxied household income measure.[10] In Online Table A.9, we use an alternative proxy for present bias constructed by regressing each driver's daily session minutes supplied against the days-since-Monday categorical variable over the entire baseline period. This proxy similarly captures how likely a driver was to concentrate their labor closer to payday, and we similarly find that a one standard deviation increase in this measure strongly predicts higher take-up.

Third, we examine more model-driven patterns in heterogeneous treatment effects implied by present bias and liquidity constraints, respectively. In particular, drivers in the control group had to wait nine days to receive pay for work done on Monday, eight days for

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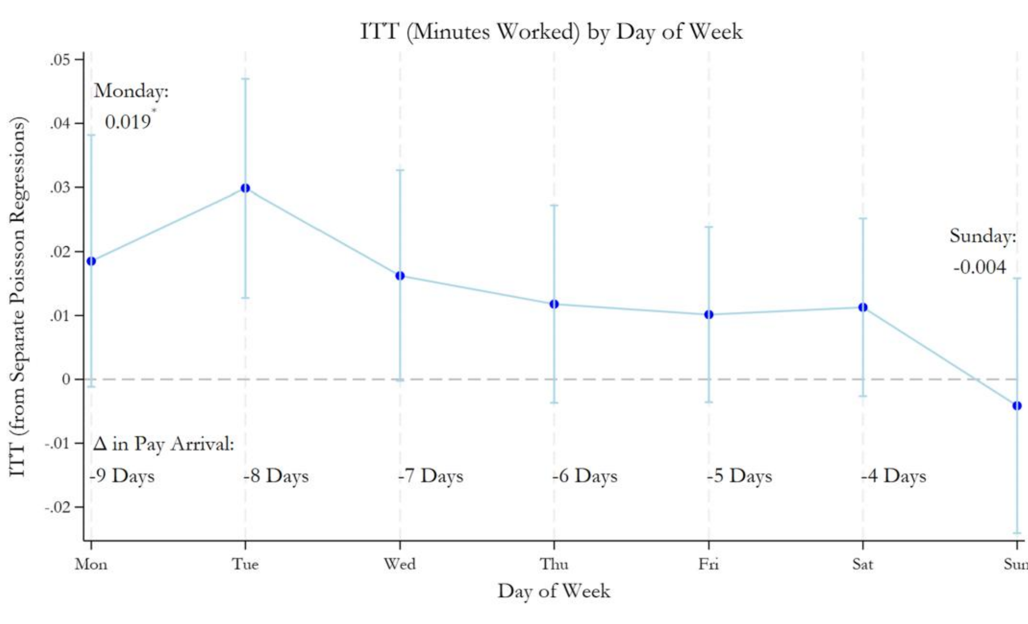
Tuesday, and so on until a gap of three days for work done on Sunday. Instant Pay shrunk this gap to zero across all days of the week. Certain formulations of present bias, such as the $1/T$ hyperbolic discounting model, would predict a behavioral response that shrinks as the counterfactual pay delay gets smaller. Figure 4 shows this pattern when we separately estimate by each day of the week with a 1.9% ITT on Monday mostly decreasing out to -0.4% ITT on Sunday (Online Figure B.2 shows that this pattern is a bit more pronounced and tightly estimated in our extended sample). By contrast, liquidity constraints and pressing financial needs suggest either no systematic pattern (if labor supply responses match idiosyncratic financial shocks) or that the pattern matches background variation in liquidity. One challenge to estimating background variation in liquidity is that Uber drivers often have multiple other sources of income, including work on other gig platforms.[11] One liquidity event that we know with certainty is the usual Uber paycheck direct deposits on Wednesday. If liquidity constraints are operative, we expect the smallest response on Wednesday when control group drivers receive their usual paycheck and for the response to increase out to Tuesday. Whereas Figure 4 does not show this pattern of increasing effects from Wednesday to Tuesday, we run a horse race through regressions in Table 3. Specifically, we interact $Treatment_{iz} * Post_{izj}$ with categorical variables counting the number of days since Monday (zero to six) or since Wednesday (zero to six). Column (2) suggests this linear specification with just the days-since-Monday interaction broadly parallels the subsample analysis, showing an ITT of 2.6% on Monday and this effect decreasing by 0.4 pp each day out. By contrast, column (3) shows that the interaction with days-since-Wednesday is insignificant (though in the direction predicted by liquidity constraints). The coefficient, however, shrinks closer to zero in column (4) when both interactions are estimated with only the days-since-Monday interaction remaining significant in panel A. Finally, in Online Appendix C, we estimate labor supply elasticities in response to randomized wage rates in another experiment by day of the week and do not find the downward-sloping pattern found for Instant Pay. This result suggests that the observed temporal pattern in response to Instant Pay is unique and does not simply reflect drivers being more elastic early in the work week in response to any incentives.

Finally, we examine the daily probability of Instant Pay cash outs and average daily amounts over time among drivers capable of cashing out. Figure 5 shows a strong weekly pattern emerging by the end of March: cash outs were the least likely and lowest in average value on Mondays and grow increasingly likely and greater in value throughout the week before peaking on Sundays. This increasing weekly pattern suggests two possibilities. First, drivers may have present-biased preferences over cash itself (rather than simply consumption), a possibility experimentally demonstrated by Andreoni et al. (2018) using electronic bank transfers. As noted by Andreoni et al. (2018), there is also physiological evidence that the receipt of cash acts the same as a primary reward (Löckenhoff et al. 2011; Lempert et al. 2015, 2016). Under this account, drivers

Figure 4. (Color online) ITT by Day of Week



Notes. Each dot corresponds to a coefficient on the interaction term (*Treat* × *Post*) from a separate Poisson regression for each "Day of the Week" subsample. The bars correspond to the 95% confidence interval (±1.96*SE) on that point estimate. Regressions include fixed effects for the interactions of randomization cohort and date. Standard errors are clustered by driver.

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Table 3. Days-Since-Monday vs. Days-Since-Wednesday

	Minutes			
	(1)	(2)	(3)	(4)
Treat × Post	0.014**	0.026***	0.009	0.023**
	(0.006)	(0.009)	(0.007)	(0.010)
Treat × Post × Days Since Monday		−0.004**		−0.004**
		(0.002)		(0.002)
Treated × Post × Days Since Wednesday			0.002	0.001
			(0.001)	(0.001)
N	13,242,030	13,242,030	13,242,030	13,242,030
DepVarMean	121.752	121.752	121.752	121.752

Notes. Column (1) repeats the specification used in Table 1 to estimate the ITT on minutes worked (panel A), earnings (panel B), the probability of working (panel C), and minutes conditional on working (panel D). Column (2) modifies this by including an interaction with Days-Since-Monday which goes from zero (Monday) to six (Sunday), whereas column (3) instead does so for Days-Since-Wednesday from zero (Wednesday) to six (Tuesday). Column (4) reports a specification with both interactions included. Specifically, all specifications are from Poisson regressions with fixed effects for the interactions of randomization cohort (city-by-enrollment-date) and date. DepVarMean is the mean in the control group in the preperiod. Standard errors are clustered by driver.

*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

may be increasing labor supply in response to the now immediate option value of accessing their earnings even if they do not have immediate consumption needs. As a result, patterns in labor supply may be completely decoupled from patterns in cashing out. Second, to understand why cash outs increase out to Sunday rather than just being idiosyncratic, recall that the deadline to cash out falls on that date. Drivers may have procrastinated pushing the button within the driver app to transfer their earnings until Sunday night, after which any remaining funds were unavailable until Wednesday.[12] In summary, if liquidity constraints and immediate consumption needs were the only mechanism driving the response to the EWA, we would expect the patterns of labor supply and cash outs to move in lockstep. Instead, we see exactly opposite trends in driving and Instant Pay use over the course of a week.

In sum, the evidence presented in this section suggests that present bias over access to cash provides a parsimonious explanation for many of the patterns observed in the data. Our results on the relevance of time preferences for explaining the labor supply response to Instant Pay are also consistent with findings on the consumption side by De La Rosa and Tully (2022), who find a correlation between intertemporal discount rates and increases in consumption among laboratory participants given access to an EWA product. However, to be clear, several mechanisms could be present simultaneously. For example, Smith (2022, p. 68), in an ethnography of Los Angeles rideshare drivers between 2018 to 2020, finds that liquidity-constrained drivers often struggle with the asynchronous timing of pay and expenses and use "a range of techniques, apps, and obstacles both mental and physical to help themselves split or aggregate their income and expenses, so that they have the right amount of money at the right time." He notes the varied ways that drivers find Instant Pay helpful for managing these challenges. Moreover, there are some other psychological accounts that could be relevant. For example, Smith (2022, p. 66) discusses gamification; whereas the driver app already displayed earnings after each trip and weekly cumulative totals, EWA products enabled drivers to "start at $0 each day so that [they] can easily track [their] progress." In other words, coupling the usual daily-level feedback with the immediate option-value reward could increase intrinsic work motivations (Woolley and Fishbach 2018, Liu et al. 2022). It is also possible that the separate Instant Pay account either allowed drivers to segment earnings in a way that provided a higher effective savings rate (either because of mental accounting or actual accounting in the intrahousehold bargaining we speculate may drive the sex correlation). Whereas these additional accounts may play some role in the overall labor supply effects, they do not predict the observed day-of-the-week patterns. Altogether, whereas our evidence is indirect, it collectively suggests an important role for present bias.

7. Conclusion

Using a nationwide experiment at Uber, this paper shows that allowing relatively instant access to accumulated earnings can increase the labor supply of workers. Whereas previous work documents how variation in pay frequency can affect consumption in contexts with limited work flexibility, this paper instead focuses on labor supply responses to introducing flexible pay in a context with flexible work. Although standard models suggest that changing pay frequency from weekly to on-demand should have little effect, we find that the labor supply response is on par with increasing

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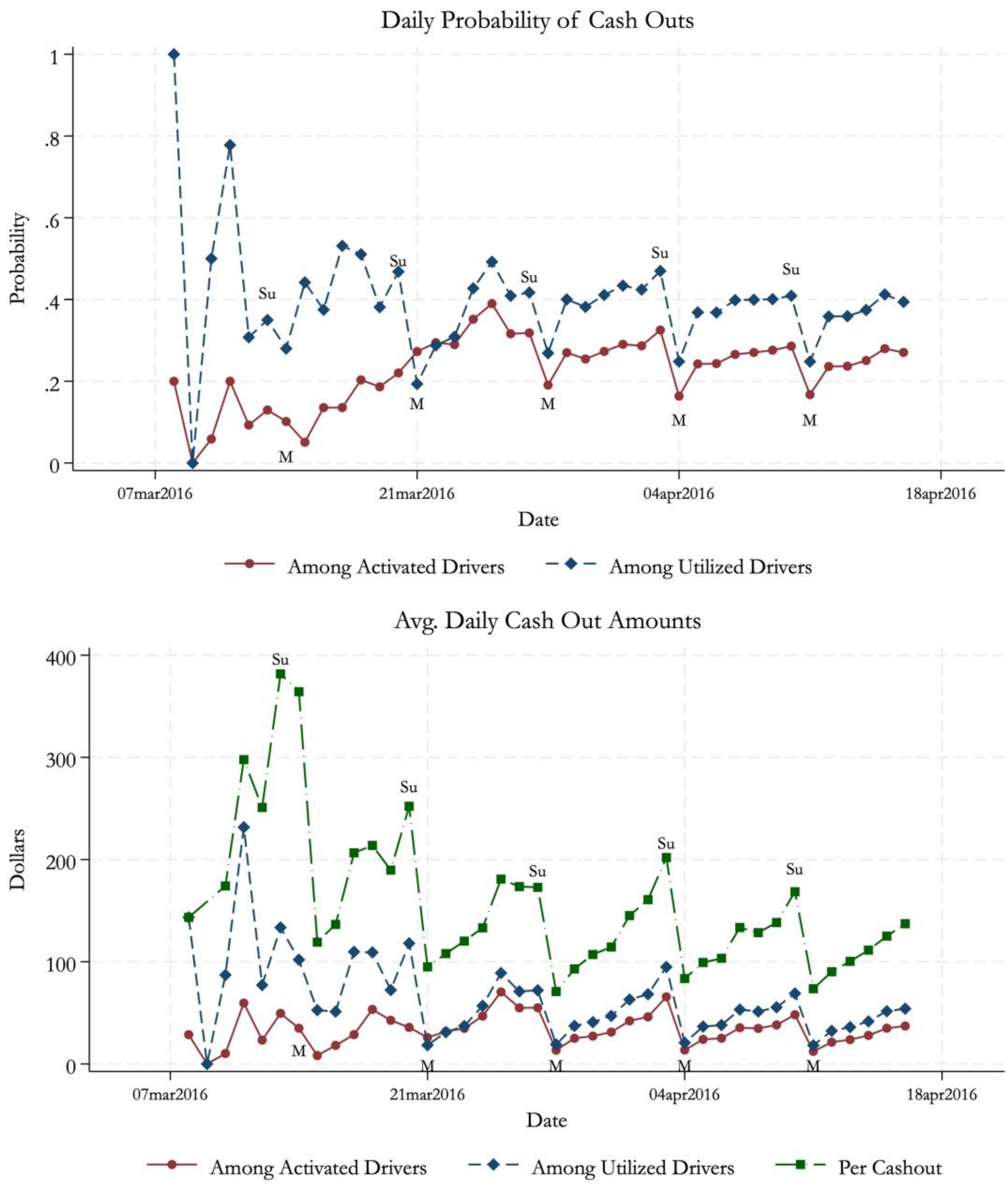
Figure 5. (Color online) Daily Cash Out Probability and Average Amounts



Notes. This figure plots the daily probability of cashing out earnings using Instant Pay and average amounts. Daily likelihoods and amounts cashed out among activated and utilized drivers are restricted to the groups of drivers capable of cashing out on that day (e.g., March 28 is the first date on which there were at least 1,000 drivers who had activated their cards). We say that a driver is capable of initiating a cash out on a given day if the driver has either activated an Uber debit card or cashed out (utilized) at any point up to and including that day. By contrast, the average daily amount per cash out is the total dollars cashed out divided by the number of cash outs on that day.

wages by 11%. Moreover, the increase in work hours is concentrated in days further away from the counterfactual payday in line with previous research documenting payday effects in labor supply (Kaur et al. 2015).

Estimating the welfare and possible general equilibrium effects of Instant Pay is beyond the scope of this paper; however, we discuss some potential channels through which both consumer and driver welfare may be affected. On the consumer side, by increasing the supply of drivers at any given moment, the policy may lower wait times and slightly depress both the optimal base fare and surge multipliers, all of which should increase consumer welfare. Whereas we do not see evidence of significant price changes, such price changes could have occurred after the feature was available to all drivers. Second, increasing pay frequency may improve ride quality for consumers by reducing the psychological effects of financial concerns for drivers (e.g., improving attention and focus as in Kaur et al. 2025),

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especially if they respond to the option value rather than actual receipt of early wages.[13]

The effects of Instant Pay on driver welfare largely depend on whether drivers are sophisticated or naïve about their time preferences. If drivers do not suffer from present bias, then Instant Pay may increase welfare by relaxing very tightly binding liquidity constraints and by allowing them to withdraw pay at a cadence better matching their expenses. If drivers do display present bias and are sophisticated (i.e., they correctly forecast their future periods' labor–leisure trade-offs), then they will only enroll in Instant Pay if it is welfare improving because enrollment has a hassle cost and only facilitates early access to payment after getting the card (or, in some limited cases, an ACH transfer) in a future period. Thus, if infrequent (weekly) pay serves as a commitment device to save or limit temptation spending, sophisticates would only undo it by opting into Instant Pay if they correctly anticipate that the additional incentive to work would fund the increase in consumption in a way that at least weakly increases present discounted expected utility.[14]

There is, however, ambiguity on welfare in the case of full or partial naivete.[15] A naïve driver may enroll in the program to alleviate an anticipated future midweek expense but fail to realize that the additional liquidity creates a permanent increase in temptation. On the other hand, drivers may also underestimate how this future temptation to spend will increase their incentive to work. This second biased belief about work incentives could be enough to outweigh the distortion created by the self-control problem over consumption.[16] Tempering these concerns, recall that drivers' day-of-the-week cash out patterns do not match the high early-in-the-week pattern displayed by the additional trips induced by Instant Pay, a synchronicity we would have expected if money earned through Instant Pay went predominantly to finance immediate temptation spending.

Additionally, whereas we focus on present bias and beliefs about present bias (naivete), it is possible that workers also misforecast expenses (e.g., suffer from budget neglect) and that faster access to pay exacerbates overspending. Whereas this concern may be somewhat muted in considering the switch from weekly to daily pay, it could be relevant for larger shifts in pay frequency and may motivate pairing such shifts with planning aids (Augenblick et al. 2022). Finally, the welfare effects all of these considerations likely depend on the potential of Instant Pay to substitute for high-interest vehicles such as payday loans and the welfare impacts of reductions in the utilization of high-cost debt (Allcott et al. 2022).

This paper shows that (a) there is demand for increased pay flexibility even among weekly paid workers and (b) increased pay flexibility can benefit the firm by increasing labor supply through a relatively inexpensive lever. As such, firms are likely to increasingly offer such pay flexibility, particularly in contexts in which the firm benefits from increased labor supply. Whereas our limited observation window precludes an analysis of pay flexibility on Uber driver retention, other research suggests that such flexibility may also benefit firms through lower employee turnover (Baker 2018, Murillo et al. 2023).

This shift toward greater pay flexibility is growing as more financial companies provide EWA solutions to companies and workers directly. For example, the largest U.S. employer, Walmart, allows workers to receive up to 50% of their pay early once per week via the similarly named Instapay, facilitated by the financial application Even (Corkery 2017). This service was utilized by more than 200,000 workers just eight months after the 2017 launch (Crosman 2018), and Even also provided financial planning and savings tools that may help address some of the consumption misforecasting risks of higher pay frequency though more research is needed on possible consumption effects (Crosman 2018). Similar EWAs are also being offered and utilized by employees of McDonalds, Outback Steakhouse, DoorDash, and GrubHub, among others (Gee 2017), and in 2022, more than seven million U.S. workers used EWAs to access around $22 billion in earnings (Consumer Finance Protection Bureau 2024).

It is important to note several risks associated with the use and growth of EWA products as identified by the Consumer Finance Protection Bureau (CFPB). Workers can face relatively high transaction fees, especially for expedited payments, under the direct-to-consumer model or under the employer-integrated model when the employer does not subsidize such fees. These high fees can be equivalent to annual percentage rates of more than 100% and place additional financial burden on frequent EWA users. The CFPB also raises concerns that workers may financially overextend themselves if they use multiple EWA products simultaneously, increasing the likelihood of incurring additional overdraft or nonsufficient funds fees. Finally, workers' limited understanding of the terms and conditions of these products, especially because the EWA products are not credit and lack regulatory oversight, may also lead to unintentional misuse and unexpected fees, potentially increasing the financial instability of EWA users (Consumer Finance Protection Bureau 2024). Finally, whereas Instant Pay was free to use during the experiment and for several years after, that is no longer the case for most drivers, so caution should be taken in extrapolating to the present version of the feature.

Whereas our paper provides causal evidence on the labor supply margin, future work may benefit from pairing a randomized controlled trial (RCT) with

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administrative data on labor supply as well as spending and borrowing to better understand the full financial impact of these policies. Moreover, given previous work documenting demand for commitment in the form of deferred pay in developing country contexts (Casaburi and Macchiavello 2019, Brune et al. 2021), there may be value to offering workers the option to commit to pay deferral as part of the menu of pay contracts.

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Endnotes

[1] See Ericson and Laibson (2019) for a broad overview of models described under an umbrella of "present focus" intended to remove the prejudgment that the behavior is a mistake.

[2] Baugh and Wang (2022) find a higher likelihood of financial shortfalls (reflected in bank overdrafts, bounced checks, and online payday lending usage) among social security benefit recipients with longer pay cycles (five versus four weeks) but similar due dates for recurring bills; they argue these results are best explained by reliance on simplistic budgeting heuristics. By contrast, Baugh and Correia (2022) examine cross-sectional variation in pay frequency in an account aggregator; they find that those paid at a higher frequency have more financial shortfalls despite lowering credit card borrowing and consumption. They argue that this result can be explained by extending a model of optimal pay timing for present-biased workers (Parsons and Van Wesep 2013) with a costless borrowing option and an illiquid savings vehicle; theoretically, the higher liquidity of frequent pay should increase the likelihood of investing in the illiquid savings vehicle, which would help explain the other three empirical results. Finally, De La Rosa and Tully (2022) find that higher pay frequency is correlated with increased spending among clients of a financial services provider and, in lab studies, find evidence that higher pay frequency increases subjective wealth perceptions (but that such effects depend on the timing of expenses).

[3] As described in Chen et al. (2019), as ride requests emerge, Uber's algorithm assigns these to drivers based on proximity. At the time of our study, riders were charged a base fare along with additional costs determined by the distance and duration of the trip. Fares were standardized within each city and subject to dynamic pricing during periods of high demand relative to driver availability in specific areas. Excluding various taxes, fees, and promotional adjustments, drivers received a portion of the total fare less Uber's service fee (roughly 20% to 30% during the study period of 2016). Thus, a driver's income is influenced by their choices regarding work hours and location as well as the fluctuating demand and availability of other drivers and riders.

[4] Online Table A.6 reproduces our primary regression results, including drivers assigned to Minneapolis–St. Paul, Minnesota, and shows an increase in the magnitudes and statistical significance of our treatment effects across all outcomes. The failed random assignment in this city led to meaningful baseline differences across the two groups. Over the two weeks prior to the experiment, the average daily driving minutes were 44 minutes higher for treatment than control drivers, and the difference in average daily earnings was $15.46. After careful consideration, we chose to drop this city and its drivers to avoid biasing our treatment effects and potentially overstating the causal effects of the program.

[5] Driver experience in months is computed by taking the difference between April 16, 2016, and the date on which a driver was onboarded to Uber (and, thus, may span long breaks). We are missing coverage on demographics for about 6% of drivers.

[6] The coefficient may also vary if there are heterogeneous treatment effects by time since gaining access. For example, if there is a novelty effect, then we may expect larger effects in the earlier days of the window, counteracting the possible attenuation from defining the *Post* window earlier. Conversely, if drivers gradually ramp up their usage of the card, this would produce larger coefficients the further out the *Post* window date is defined.

[7] Limiting to the 55,841 treated drivers who were enrolled at least 22 days prior to April 16, we observe them roughly 32 days after enrollment, on average, with the 10th percentile being 29 days and the 90th being 37 days. As noted below, we focus on treatment effects 21 days after enrollment, so our treatment effects are identified off 12 known cities with experiment start dates at least three weeks before the April 16 cutoff and a handful of drivers in the catch-all synthetic city (498). This group of early cities consists of 111,503 drivers with 55,841 assigned to treatment and 55,662 to control.

[8] The most salient concern is that the labor supply elasticities reflect substitution away from Lyft. We cannot identify whether any drivers in our sample drive for Lyft; however, Koustas (2018), using data from a financial aggregator spanning 18,000 rideshare drivers from 2012 to 2016, finds that, "conditional on being an Uber driver, 93.3 percent of rideshare earnings come from Uber. Conditional on being a Lyft driver, 33 percent of earnings come from Uber." Thus, the scope of Lyft to Uber substitution may be somewhat limited.

[9] Gelman (2022) uses linked consumption and liquid asset data paired with a buffer stock model of consumption to implement a Euler equation test. He concludes that excess sensitivity of consumption to income (paychecks) is ultimately driven by quasi-hyperbolic discounting (though liquidity constraints, themselves driven by preferences, are a proximate cause).

[10] We can only speculate, but the strong relationship between sex and take-up may reflect the value of the Uber debit card providing a separate account for earnings. For example, if women share their primary direct deposit account with a spouse, then Instant Pay allows them to transfer some of their earnings toward a separate account over which they may have more control.

[11] During our sample period, DoorDash drivers were also paid on Wednesdays, whereas Lyft drivers could receive their weekly direct deposits any day between Tuesday and Sunday, depending on their banks' processing times (DoorDash 2016, Lyft 2016b). Additionally, drivers working for Lyft with access to Express Pay could cash out their Lyft earnings at any point after earning at least $50 that week (Lyft 2016a).

[12] One concern with looking at cash-out probabilities by day of the week is that drivers may have a heuristic of only cashing out once their accumulated earnings have reached a particular threshold. Whereas there is no fee to cash out, it is possible that the small effort of pushing the cash-out button is enough friction to preserve such a rule. Because accumulated earnings are automatically sent for direct deposit at the end of the weekly earnings cycle, accumulated earnings

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are mechanically small on Mondays, and thus, this heuristic on its own could produce the observed pattern. To imperfectly account for this, in Online Table A.10 and Online Figure A.5, we reproduce this daily probability figure, controlling for accumulated earnings prior to any cash outs and show that the pattern is virtually unchanged, suggesting that this pattern is not simply reflecting differences in accumulated earnings by day of the week.

[13] Kaur et al. (2025) test the effects of varying pay timing, holding total pay constant; specifically, they employ Indian factory workers for a two-week contract and randomize some to be paid part of the total amount four days early. They find that these treated workers exhibit a 7% increase in output in the final four days and find evidence that this is through fewer costly mistakes by alleviating the psychological effects of financial strain. However, whether these financial strain effects would be relevant among Uber drivers in the United States is unclear. As they note, "The relative liquidity boost from being paid daily or monthly would be different in long-run employment—indeed, having such stable employment would limit the likelihood that a worker faces large financial strain in the first place (e.g., Morduch and Schneider 2017)."

[14] A recent literature studies worker preferences over pay timing. Parsons and Van Wesep (2013) develop a model of optimal pay by firms with present-biased workers with the result that a sophisticated worker would be willing to accept a lower average wage in exchange for a contract that better matches pay timing with regular consumption needs (because the worker's future self would not make the savings decisions that the worker's current self prefers). See also Casaburi and Macchiavello (2019) and Brune et al. (2021) for RCTs on less frequent or deferred pay.

[15] Kaur et al. (2025) make a similar point, "Suppose that a worker is paid monthly and also has rent due monthly. If that worker receives a weekly payment, self-control problems may lead them to save too little and at the end of the month they may not be able to make rent payments. Weekly payment may—when combined with lumpy consumption and imperfect self-control—create more financial strain."

[16] The exact sign on this depends on the shape of the effort cost function and slack in the work hours constraint. For example, for naïve present-biased drivers who cannot work any additional hours, Instant Pay increases the temptation to spend without being able to adjust the work margin. Whether the additional benefits of Instant Pay (alleviating some financial strain, allowing the worker choice over the payday) outweigh this cost is unclear in this case. Future research pairing this variation with detailed consumption and liquid asset data—paired with a structural model—would allow a more complete treatment.

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